Email Security Gateway Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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Contents

1 Introduction 2
- 1.1 Purpose . . . 2
- 1.2 Scope . . . 2
- 1.3 Definitions, acronyms and abbreviations . . . 2
- 1.4 References . . . 2
- 1.5 Overview . . . 2

2 Overall description 2
- 2.1 Product perspective . . . 2
- 2.2 Product functions . . . 2
- 2.3 User characteristics and stakeholders . . . 3
- 2.4 Constraints . . . 3
- 2.5 Assumptions and dependencies . . . 3

3 Specific requirements 4
- 3.1 External interface requirements . . . 4
  - 3.1.1 User interfaces . . . 4
  - 3.1.2 Hardware interfaces . . . 4
  - 3.1.3 Software interfaces . . . 4
  - 3.1.4 Communications interfaces . . . 4
- 3.2 Functional requirements . . . 4
- 3.3 Performance requirements . . . 5
- 3.4 Logical database requirements . . . 5
- 3.5 Design constraints . . . 5
- 3.6 Software system attributes . . . 5
  - 3.6.1 Reliability . . . 5
  - 3.6.2 Availability . . . 5
  - 3.6.3 Security . . . 5
  - 3.6.4 Maintainability . . . 6
  - 3.6.5 Portability . . . 6
- 3.7 Other requirements . . . 6

1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links. The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages. The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events. The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures and the SDLC plan are available in the supporting report.

2 Overall description

2.1 Product perspective

Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- Access permissions — FR-006, FR-007
- DNS and web filtering — FR-008
- Monitoring and alerts — FR-005, FR-010
- Dashboard — FR-010
- Audit logging — FR-008
- protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links — FR-001
- inspect incoming and outgoing email traffic — FR-002
- detect suspicious messages — FR-003
- quarantine potentially harmful emails — FR-004
- integrate with the university identity system — FR-009

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
User	students, faculty, staff	Uses permitted functions; each source class keeps its specified permissions.
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- What conditions define suspicious email content?
- Which fields must the audit logs contain?
- What retention period is required for the audit logs?
- Which security acceptance outcomes must be demonstrated?
- What measurable reliability and recovery targets must the system meet?
- What measurable scalability targets must the system meet?
- What criteria define acceptable manageability?
- What conditions permit quarantining and releasing messages?
- Will the system implement micro-segmentation to isolate email traffic from other network segments?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-010. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: FR-009. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall inspect incoming and outgoing email traffic.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall detect suspicious messages.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall quarantine potentially harmful emails.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall generate security alerts.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall allow authorized administrators to review quarantined messages.

Traceability: E-002, TEST-006

FR-007 · Should · explicit

The system shall allow authorized administrators to release quarantined messages.

Traceability: E-002, TEST-007

FR-008 · Should · explicit

The system shall maintain audit logs of filtering decisions and administrator actions.

Traceability: E-003, TEST-008

FR-009 · Should · explicit

The system shall integrate with the university identity system.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall provide a dashboard for monitoring email security events.

Traceability: E-003, TEST-010

3.3 Performance requirements

NFR-004 · Should · explicit

The system shall be scalable for university-wide deployment.

Traceability: E-004, TEST-014

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

NFR-003 · Should · explicit

The system shall be reliable for university-wide deployment.

Traceability: E-004, TEST-013

NFR-005 · Should · explicit

The system shall be manageable for university-wide deployment.

Traceability: E-004, TEST-015

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-005, FR-008, FR-010.

NFR-001 · Must · explicit

The system shall be secure.

Traceability: E-001, TEST-011

NFR-002 · Should · explicit

The system shall be secure for university-wide deployment.

Traceability: E-004, TEST-012

3.6.4 Maintainability

Also covered by: NFR-005.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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